

Speech-to-Text Quality & Accuracy

Two independent evaluations of the faster-whisper `large-v3` transcription output for the persona-clone corpus: how accurate it is against real reference text, and how internally consistent it is across every file.

CORPUS 199 files, Sadhguru talks & interviews MODEL faster-whisper large-v3

HARDWARE NVIDIA T1000 GENERATED 2026-08-06

SECTION 1 Model Accuracy Evaluation

Measures whether Whisper transcribed the *actual spoken words* correctly — Word Error Rate, Character Error Rate, Match Error Rate, and Word Information Lost, computed with `jiwer` against YouTube's manually-uploaded (non-auto-generated) English subtitles as ground truth. **122 of 199** videos had a manual subtitle track available; the remaining **77** are not scored here, as no independent reference exists for them.

WER	CER	MER	WIL
7.23%	4.67%	7.00%	9.51%
median 6.02% · max 24.08%	median 3.38% · max 24.71%	median 5.93% · max 23.36%	median 8.29% · max 34.51%

A mean WER of **7.2%** on real, unscripted conversational speech is a strong result for this model class — comparable benchmarks on clean studio speech score 2–3%, and 5–15% is the expected range for informal dialogue like interviews and talks.

p10-p90 spread	median	max (worst file)	
WER			7.2% mean
CER			4.7% mean

Scale: 0–40% WER/MER, 0–40% CER, 0–40% WIL (tick positions shown to scale within each row).

WORST 5 FILES BY WER

VIDEO ID	WER	CER	MER	WIL	REF WORDS	HYP WORDS
IzLIIfzj6xfo	24.08%	10.27%	22.34%	34.51%	1,119	1,195
bzx0nynHIzs	23.92%	24.71%	23.36%	25.11%	887	727
F5378Ag9EjA	23.78%	11.56%	21.54%	31.84%	722	793
32brcqoNa7Q	23.08%	18.09%	21.77%	33.92%	8,813	9,123
4Fl_Dxk4Z7s	20.49%	18.56%	20.20%	22.93%	497	419

bzx0nynHIzs is the one anomaly worth a closer look: its CER (24.7%) actually exceeds its WER (23.9%), and its word count is 18% short of the reference (727 vs 887) — a pattern more consistent with a chunk of dropped/missing audio than ordinary word substitutions. The other four worst files show hypothesis word counts close to or above the reference, typical of normal substitution-driven error.

BEST 5 FILES BY WER

VIDEO ID	WER	CER	MER	WIL	REF WORDS	HYP WORDS
RjXCSSLktkI	0.38%	0.20%	0.38%	0.57%	530	529
diFkCJ802vY	1.62%	0.73%	1.61%	1.97%	561	560
SAnUTDJ-DIY	1.94%	1.38%	1.94%	2.26%	315	309
tMKuQZS6GBU	1.97%	0.56%	1.97%	2.82%	813	807
oMoNZZoCJIY	2.01%	1.33%	2.00%	3.23%	403	401

SECTION 2Transcription Quality Evaluation

Measures the transcription itself for internal consistency and reliability — not correctness against ground truth, since only 61% of the corpus has one. Built from signals faster

against ground truth, since only 0.1% of the corpus has one. Built from signals faster-whisper produces natively (per-word confidence) plus structural heuristics (repetition, speech coverage, pacing), so it runs across the full 199-file corpus.

AVG WORD CONFIDENCE	LOW-CONFIDENCE WORDS	PACE	SPEECH COVERAGE
0.948	3.2%	2.09	75.5%
median 0.950 · min 0.837	median 3.0% · max 16.6%	words/sec · median 2.09	of audio duration, median 75.9%
REPETITION SCORE	FILES FLAGGED		
0.034	1 / 199		
median 0.029 · max 0.112	for manual review		

p10–median spread	median	min (weakest file)
Confidence	0.948 mean	
Pace (wps)	2.09 mean	
Coverage	75.5% mean	

Scale: Confidence & Coverage 0–100%; Pace 0–5 words/sec. Min tick shown instead of max here – lower is the direction of concern for these three.

FILES FLAGGED FOR MANUAL REVIEW

VIDEO ID	DURATION	AVG CONFIDENCE	LOW-CONF WORDS	REASON
HRXrfIvRzNY	4:44	0.837	16.6%	Many low-confidence words

Only one file out of 199 was flagged, and even it isn’t severe — 0.837 average confidence versus a corpus mean of 0.948, and no repetition or coverage problems. This corpus shows no sign of the systemic failure modes (hallucination loops, silence misread as speech, dropped audio) that Whisper-family models are known to occasionally produce.

SUMMARY

Reading the two together

The two evaluations measure different things — correctness against real reference text versus the model's own internal consistency — and they agree. The one file flagged for low confidence (HRXr~~f~~IvRzNY) has no manual subtitle available, so its true WER is unknown, but its confidence profile is the corpus's weakest point either way. Nothing in either evaluation suggests systemic transcription problems; this corpus is ready to move into diarization.

Methodology & sources

- Model accuracy: `eval/asr/fetch_references.py` (YouTube manual subtitles via yt-dlp) → `eval/asr/compute_wer.py` (jiwer, lowercase + punctuation-stripped before scoring) → `eval/results/asr_accuracy_report.json`
- Transcription quality: `src/transcription/quality.py` → `src/transcription/evaluate.py` → `eval/results/transcription_quality_report.json`
- Reference text excludes speaker labels and parenthetical stage directions (e.g. "(Laughs)"), stripped by `eval/asr/vtt_parser.py`